```
import cv2
import numpy as np
from google.colab import files
from matplotlib import pyplot as plt

# Upload image
uploaded = files.upload()

# Get uploaded image name
filename = list(uploaded.keys())[0]

# Read image
img = cv2.imread(filename)

# Check if image is loaded
if img is None:
    print("Error: Image could not be loaded.")
else:
    # Laplacian mask with negative center coefficient
    kernel = np.array([
        [0, 1, 0],
        [1, -4, 1],
        [0, 1, 0]
    ], dtype=np.float32)

    # Apply Laplacian mask
    laplacian = cv2.filter2D(img, cv2.CV_32F, kernel)

    # Sharpen the image
    sharpened = cv2.convertScaleAbs(
        img.astype(np.float32) - laplacian
    )

    # Save output
    cv2.imwrite("Sharpened_Image.jpg", sharpened)

    # Display original and sharpened images
```

```
    plt.figure(figsize=(10, 5))

    plt.subplot(1, 2, 1)
    plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
    plt.title("Original Image")
    plt.axis("off")

    plt.subplot(1, 2, 2)
    plt.imshow(cv2.cvtColor(sharpened, cv2.COLOR_BGR2RGB))
    plt.title("Sharpened Image")
    plt.axis("off")

    plt.show()

    # Download output
    files.download("Sharpened_Image.jpg")
```

Choose Files CVVV2.png

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```
Saving CVVV2.png to CVVV2.png
```

Original Image

Sharpened Image